

Fake News Detection Using Optimized Deep learning model through effective feature extraction.

Arati Chabukswar,
Department of Computer Science &
Eng,
UVCE,Bangalore University,
Bengaluru,India
arati.chabukswar0106@gmail.com

P Deepa Shenoy,
Department of Computer Science &
Eng,
UVCE,Bangalore University,
Bengaluru,India
shenoypd1@gmail.com

Venugopal K R,
Department of Computer Science &
Eng,
UVCE,Bangalore University,
Bengaluru,India
venugopalkr@uvce.ac.in

Abstract— Internet plays a vital role in the propagation of data through public networks or websites due to which Fake news is produced for commercial and political benefits to misinform and fascinate readers. In order to stop the scattering of fake news, detection techniques are researched based on deep learning in natural language processing tasks. The deep learning model proposed in this study combines LSTM and Bi-directional LSTM with one-hot encoding representation for classification of fake news. The model is effectively authenticated on the political news articles dataset consisting of 240 websites (URLs), body, headline, and label. TensorFlow framework utilizes built-in Keras deep learning libraries that is having an online community on TensorFlow Kaggle repository. The outcome demonstrates the Bi-directional LSTM model's advantage over the Unidirectional LSTM model in terms of accuracy

Keywords— Fake news, Natural Language Processing (NLP), Deep Learning, Tensor flow, Keras.

I. INTRODUCTION

Fake news is one of the biggest challenges to today's society, journalism, political field, prosperity, economy, and country's security. Fake news is well defined as causing intense interest or curiosity about the news, report or information that is inaccurate and is created to gain attention, misinform, or damage a person's reputation or a company. Public is not conscious of how bogus news could affect them and their surroundings and how to overcome when such things happen. In day-to-day life, access to news sources through websites, social media, and online papers has increased and knowingly or unknowingly whether the news is fake or real, people are propagating this news leading to serious issue if not controlled in a proper manner [1],[2].

The impact of fake news can be quoted with few examples as follows. The stock market experienced a decline of \$130 billion as a result of false news story stating that an injury in an explosion by Barack Obama the US President [3]. Propagation of fake news has a popular example recognized on the internet which is spread on Facebook in the year 2016 of presidential election held in US [4].

Nowadays, people access the news through online channels, websites. Fake news creation is easier in the social network platform due to low cost, ease to use, quick transformation of data. Several companies use many social media for marketing their products due to which millions of users are influenced to share and like the falsified news at a faster rate [5].

News could be related to different categories such as politics, health, education, economy, business, entertainment, and sport. In this article, fake news related to politics is detected using NLP and DL techniques.

This paper is divided into different main sections. Section II provides work summary connected to the proposed model. Section III provides the system architecture and its explanation. Section IV has the experimental results and discussion. Lastly, Section V provides conclusion and discussion about the future works.

II. RELATED WORK

Automatic detection of false news using Long Short Term Memory - Levy Flight distribution (LSTM-LF) algorithm was proved by Dheeraj et al. (2022) [5]. The experimental results discloses that the amount of accuracy and precision is high in comparison with other approaches.

The detection of bogus news using deep learning algorithms was the focus of [6]. DL methods with pre-trained word embeddings from GloVe, Word2Vec, and fastText are worked together. Among all three models, Bidirectional - LSTM has shown high performance in terms of accuracy on all four datasets. Adamax optimizer method gains the maximum performance with lower loss rate while testing the datasets.

A fake news posture detection system is built using the news body and title. PCA (Principal Component Analysis) and chi-square are used for obtaining high-quality attributes, which are subsequently supplied to the CNN-LSTM model [7]. The popular organization Fake News Challenges provided the dataset, including 4 kinds of stances namely disagree, agree, discuss and unrelated. The accuracy and F1-score have been increased by 4% and 20%, respectively. An accuracy of 97.8% is obtained when a dimensionality reduction method called PCA is applied.

In [8], authors have used two text preprocessing techniques, three context-aware word embeddings and ten Neural Networks. Among all ten models, in combination with RCNN technique and Word2Vec Word embedding that is trained on text preprocessing technique Lemma attain in total the best results on accuracy, recall, precision as 87.56%, 88.11%, 87.55%, respectively. The accuracy of this approach could still be improved using transformers such as BERT and XLNet.

Abhinav Kumar et. al., in [9] developed an ensemble-Based Deep Learning Model by fine-tuning three dissimilar variations of BERT model such as CT-BERT

BERT-Tweet, and Roberta on COVID-19 related text data to detect the falsified news and technique is linked with DL and ML techniques. After ensemble approach applied on three dissimilar variations of the BERT model, a weighted F1-score of 99.00% is obtained. The approach is built only on English language tweets, but not on multilingual social media posts as people post tweets in their local languages.

Various challenges and research directions are given by making an extensive survey on identification of false news using different Natural Language Processing methods and Deep Learning techniques based on feature selection; use of different classifiers; features that are lexical, syntactical, and statistical in nature could rise the detection rate; use of visual data (videos and images), developing web articles in real-time etc., by authors in [10].

In [11], hybrid deep learning technique i.e., CNN-RNN is used for the false news detection using two different datasets called FAKES datasets and ISOT dataset that are obtained from a famous website called Reuters and Fake articles from different sources. An accuracy of 100% is obtained for ISOT dataset and 60% accuracy is obtained using FAKES dataset. The proposed model works better on a precise dataset, but not on generalized datasets.

An efficient deep neural network is designed in [12] by considering the features based on user's social engagement and content level features of news articles to reach significant results. An accuracy of 92.30% is obtained which could be further increased by considering more text features.

In one more research, the emphasis was on news classification using LSTM-CNN hybrid approach, user classification is done using Neural network classifier (NN) and SVM classifier. Integration of text data with other features like video, images, audio available in news and user profiles spreading them in [13].

Background Work

In [1], the authors analyze text using Natural Language Processing techniques and fake news is detected by training deep learning models built on news title and content of the news. Using regular expression, tokenization, and stop words omission, the incoming data is preprocessed. Using TF-IDF or one-hot encoding, the news title and its content are converted into numerical vectors of N-gram or sequence vectors, respectively.

With built-in Keras deep learning packages and a Tensorflow Github repository, the Tensorflow framework is utilised. The experimental study discloses that the Keras neural network model trained with N-gram or sequence vectors achieves an accuracy of 90.3%. Bidirectional LSTM technique is used to find whether the news is fake or real by collecting the data from foreign websites and newspapers, producing overall accuracy of 84% in [14].

III. PROPOSED METHOD

Binary classification method is applied to resolve problems of fake news detection. The solution to this problem is obtained by finding whether the data existing in

the news article is true based on computing the bias of a news article and the relationship between the headline of news article, body and label is analyzed.

The architecture of proposed model for fake news detection is as shown in Fig. 1. It is divided into three phases: feature extraction, data pre-processing, and classification. In first phase, input data is pre-processed using tokenization, stop-words omission and Lemmatization. In the Feature extraction phase, the features are extracted using One-Hot Encoding Representation to improve accuracy and to diminish high-dimensional data. The categorization step then receives the features for use where Unidirectional and Bi-directional LSTM model is applied to categorize the news article as false or true. In the section that follows, each appropriate phase is fully described.

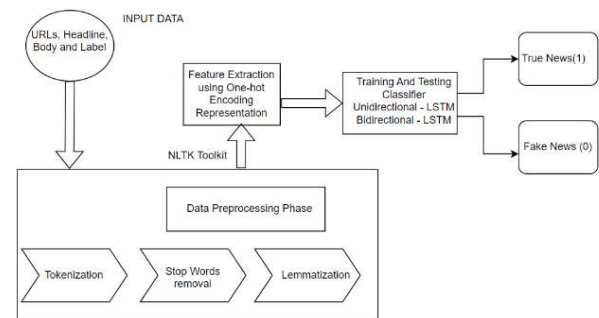


Fig. 1 Proposed model architecture.

A. Datasets for Fake News Detection

The news article in the dataset contains political datasets of websites such as www.reuters.com, www.cnn.com, www.nytimes.com, www.bbc.com, abcnews.go.com, etc.; headline, body, and binary labels as 1 for true and 0 for fake news articles. A popular non-profit website in the United States called PolitiFact is used to check the facts of political claims and reports [15].

During model evaluation on given dataset, it is divided in 80:20 ratio and comprises of true and fake news in training and testing correspondingly. It comprises of 3988 unique tokens and a count of 5000-word vectors in one-hot encoding representation. Dataset comprises of 4009 rows and 4 columns and each column describes as follows

- URLs: URL of news
- Headline: News headline
- Body: News report
- Label: 0 for fake and 1 for true news

B. Data pre-processing or text cleaning

It is a procedure that helps in converting uneven and partial data into the pattern that is understandable by machine. It includes the normal text conversion into small letter, Tokenization, stop words omission, Lemmatization using NLTK toolkit which is an open-source NLP package helps in eliminating punctuation marks, emojis, html tags, and URLs and is required to reduce dimensionality and improve model performance.

Below is a description of each step in data pre-processing.

2) *Removal of stop words*: Stop words are insignificant and not much required. Few examples are what, on, am, that, when, the etc. So, such stop words are deleted to save space and processing time.

4) *Data Visualizations*: The dataset structure is understood by data visualization. Data is visualized using Word-Cloud image for both real and fake news as shown in Fig. 2 and Fig. 3 respectively. The most frequent and significant terms in authentic and false news items are clearly highlighted in word clouds for actual news and false information.



C. Feature Extraction

Word clouds and one-hot encoding representation work together to reveal the general linguistic trends in the dataset as well as the inclusion or lack of particular words in authentic and false news tweets. These findings can assist in the creation of classification models or techniques to identify false information by enabling us to comprehend which words or keywords are typical of each group.

LSTM network can study long-term dependencies and is operative for text data because every word is linked to the one before and the one after. LSTM layer permits the model to concentrate on few parts in a sequence which are

With 10 training epochs, the model is trained on the training data-set. To prevent over-fitting, LSTM is used to news items with a dropout rate of 0.3, 100 LSTM neurons in the layer, and 1 hidden layer. The "Adam" optimizer with "binarycrossentropy" loss is employed for training the network. The default values for each of these parameters are utilised. The training data-set for the model has a batch size of 64.

Bi-directional LSTMs are the extension of old -style LSTMs used to progress the performance of model on sequence classification problems. It comprises of two LSTM components: left to right input sequence and right to left input sequence. Initial LSTM component runs on forward input sequence and other LSTM component runs on backward input sequence both running in parallel. The bidirectional LSTM has a concealed state at timestep t which is used in combination of forward and backward concealed states [14].

In this section, results on data preprocessing and model training are shown in tabular form, graphical form and using confusion matrix.

After collecting the political dataset, finally the news dataset consists of four attributes such as URLs, Headline, Body, and Label as shown in TABLE I.

ID	URLs	Headline	Body	Label
0	http://www.bbc.com/news/world-us-canada-414191	Four ways Bob Corker skewered Donald Trump	Image copyright Getty Images\n On Sunday mornin...	1
1	https://www.reuters.com/article/us-filmfestiva	Linklater's war veteran comedy speakers to moder...	LONDON (Reuters)- "Last Flag Flying", a comed..	1
2	https://www.nytimes.com/2017/10/09/us/politics..	Trump's Fight With Corker Jeopardizes His Legi...	The feud broke into public view last week when...	1
3	http://www.cnn.com/videos/cnnmonetv/2017/10/08/...	Jason Aldean opens 'SNL' with Vegas tribute	Country singer Jason Aldean, who was performin...	1

Before this step, only the name of the website is considered from the URLs. so that websites will be posting either fake news or real news but not both.

In this phase, the dataset label ratio is scrutinized in order to check whether the dataset is balanced or not as it may affect the model training outcome in further stage. The total occurrence of each label is visualized as shown in Fig. 4 in the ratio of Fake news to Real news is approximately 50:50.

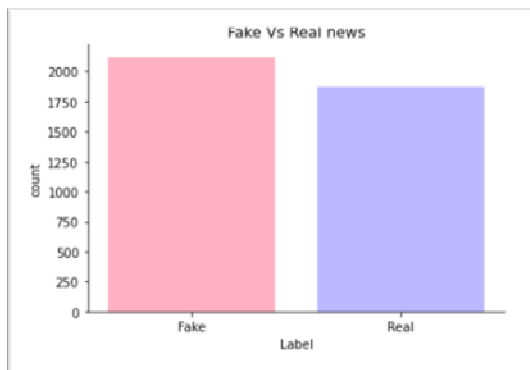


Fig. 4 Representation of Fake and Real news

Top 5 websites posting real news and fake news is plotted to find if there are any important websites in Fig. 5 and Fig. 6 respectively. It is observed that among, 250 websites, www.reuters.com website has highest count of posting real news and beforeitsnews.com website has highest count of posting fake news. This means that these two websites need to be given highest weightage while training the model.

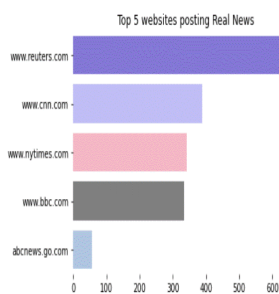


Fig. 5 Websites posting Real news

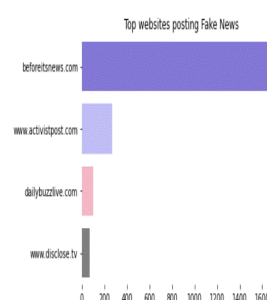


Fig. 6 Websites posting Fake news

C. Data pre-processing

In this phase, news headline and body are pre-processed using text preprocessing and vectorization method. In text pre-processing phase, regular expression, tokenization, stop words omission and lemmatization are completed on each row of news headline and news body to find the source keywords. In Fig. 7, text pre-processing is shown by taking both headline and body of news into a single column.

```
0 four way bob corker skewered donald trump imag...
1 linklater war veteran comedy speaks modern ame...
2 trump fight corker jeopardizes legislative age...
3 egypt cheiron win tie pemex mexican onshore oi...
4 jason aldean open snl vega tribute country sin...
Name: Text, dtype: object
```

Fig. 7. An example of news headline and body into one column after text pre-processing.

Next process is vectorization using one-hot encoding representation i.e., conversion of categorical data into numerical data as shown in Fig. 8

```
array([[3878, 3096, 4015, ..., 4045, 4590, 4964],
       [ 0, 0, 0, ..., 3319, 3514, 362],
       [3474, 93, 2318, ..., 4181, 3692, 3065],
       [ 0, 0, 0, ..., 4878, 3187, 4016],
       [ 0, 0, 0, ..., 582, 2954, 1915]], dtype=int32)
```

Fig. 8 An example in the form of matrix with word vectors together.

D. Model Training

To find whether the news is fake or real, classification task is required, so Keras neural network techniques with certain thick layers such as Embedding, Dropout and Dense layers are developed and trained using the TensorFlow framework on the news headlines and content in vector form. The layers 1, 3 and 5 in Keras neural network models are using the activation function named as rectified linear unit (RELU) with 64, 16 and 2 nodes respectively. Two dropout layers are used in between these 3 layers with 30% dropout rate to the network to prevent overfitting by keeping few nodes in proper place and similar work is seen in [1] where four Keras neural network models are trained with N-gram or sequence vectors on Github repository.

Sigmoid activation function is used for binary classification of Keras neural network. Training news dataset is of 67% and rest 33% is used as test set. Both models use ADAM optimizer [16] which is an optimization algorithm used to update the learning rate for each network independently.

Training on 42 samples is completed. Referring to the Fig. 9, one can see the result obtained from the Keras LSTM model held in the training stage. Model is trained with one-hot encoding (sequence vectors) representation of words. The training and validation accuracy is specified as well. The confusion matrix (provides the description of prediction results on a classification problem) of LSTM model is as shown in Fig. 10 a) depicting True negative value as 691 and True positive value as 594.

Training on 42 samples is done, the training and validation accuracy is specified. Referring to the Fig. 11 one can see the output obtained from the Keras Bi-directional LSTM model held in the training stage. Model is trained with one-hot encoding (sequence vectors) representation of words. The confusion matrix (helps in better data visualization) of Bi-directional LSTM model is as shown in Fig. 10 b) depicting True negative value as 693 and True positive value as 600.

To visually represent the findings, such as training/validation accuracy/loss curves, the graphs are as shown in Fig. 12 a) and b) for LSTM and the same is shown for Bi-LSTM in Fig.13 a) and b)

```

Epoch 1/10
42/42 [=====] - 36s 746ms/step - loss: 0.5464 - accuracy: 0.7799 - val_loss: 0.2418 - val_accuracy: 0.9058
Epoch 2/10
42/42 [=====] - 27s 635ms/step - loss: 0.1430 - accuracy: 0.9584 - val_loss: 0.1333 - val_accuracy: 0.9537
Epoch 3/10
42/42 [=====] - 28s 677ms/step - loss: 0.0514 - accuracy: 0.9839 - val_loss: 0.0772 - val_accuracy: 0.9772
Epoch 4/10
42/42 [=====] - 27s 641ms/step - loss: 0.0155 - accuracy: 0.9959 - val_loss: 0.0707 - val_accuracy: 0.9704
Epoch 5/10
42/42 [=====] - 29s 692ms/step - loss: 0.0040 - accuracy: 1.0000 - val_loss: 0.0660 - val_accuracy: 0.9795
Epoch 6/10
42/42 [=====] - 27s 640ms/step - loss: 0.0028 - accuracy: 0.9996 - val_loss: 0.0732 - val_accuracy: 0.9749
Epoch 7/10
42/42 [=====] - 27s 638ms/step - loss: 0.0089 - accuracy: 0.9985 - val_loss: 0.0863 - val_accuracy: 0.9704
Epoch 8/10
42/42 [=====] - 27s 644ms/step - loss: 0.0114 - accuracy: 0.9989 - val_loss: 0.0575 - val_accuracy: 0.9765
Epoch 9/10
42/42 [=====] - 27s 651ms/step - loss: 0.0031 - accuracy: 0.9996 - val_loss: 0.0666 - val_accuracy: 0.9810
Epoch 10/10
42/42 [=====] - 27s 642ms/step - loss: 0.0012 - accuracy: 1.0000 - val_loss: 0.0769 - val_accuracy: 0.9757

```

Fig. 9 An example output of Keras LSTM model trained.

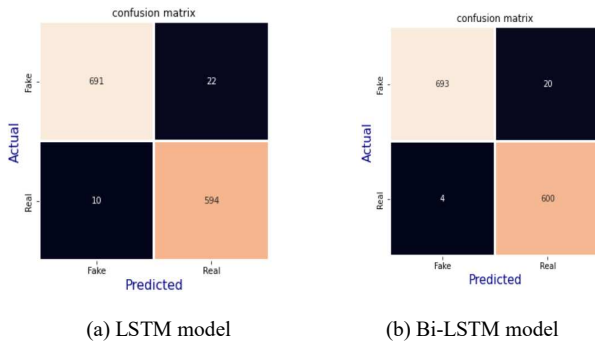


Fig. 10: Confusion matrix

```

Epoch 1/10
42/42 [=====] - 28s 665ms/step - loss: 6.8059e-04 - accuracy: 1.0000 - val_loss: 0.0767 - val_accuracy: 0.9772
Epoch 2/10
42/42 [=====] - 29s 669ms/step - loss: 5.4932e-04 - accuracy: 1.0000 - val_loss: 0.0712 - val_accuracy: 0.9787
Epoch 3/10
42/42 [=====] - 27s 647ms/step - loss: 4.0297e-04 - accuracy: 1.0000 - val_loss: 0.0782 - val_accuracy: 0.9772
Epoch 4/10
42/42 [=====] - 27s 647ms/step - loss: 3.0994e-04 - accuracy: 1.0000 - val_loss: 0.0813 - val_accuracy: 0.9787
Epoch 5/10
42/42 [=====] - 27s 652ms/step - loss: 9.9199e-04 - accuracy: 0.9996 - val_loss: 0.0912 - val_accuracy: 0.9742
Epoch 6/10
42/42 [=====] - 28s 664ms/step - loss: 0.0143 - accuracy: 0.9970 - val_loss: 0.0704 - val_accuracy: 0.9734
Epoch 7/10
42/42 [=====] - 28s 665ms/step - loss: 0.0027 - accuracy: 0.9996 - val_loss: 0.0703 - val_accuracy: 0.9757
Epoch 8/10
42/42 [=====] - 28s 673ms/step - loss: 0.0010 - accuracy: 1.0000 - val_loss: 0.0737 - val_accuracy: 0.9780
Epoch 9/10
42/42 [=====] - 28s 679ms/step - loss: 6.4856e-04 - accuracy: 1.0000 - val_loss: 0.0754 - val_accuracy: 0.9765
Epoch 10/10
42/42 [=====] - 30s 709ms/step - loss: 4.3847e-04 - accuracy: 1.0000 - val_loss: 0.0798 - val_accuracy: 0.9818

```

Fig. 11 An example output of Keras Bi-directional LSTM model trained.

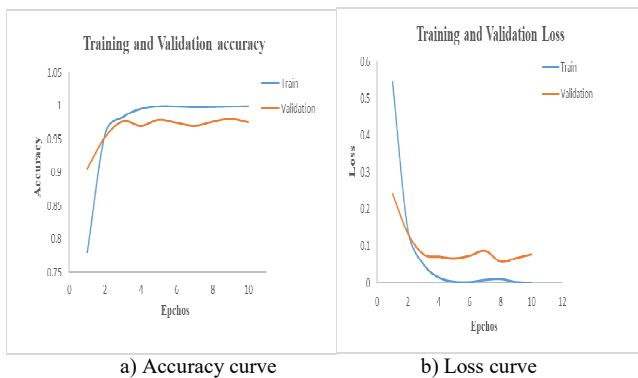


Fig. 12: LSTM Training and Validation curves

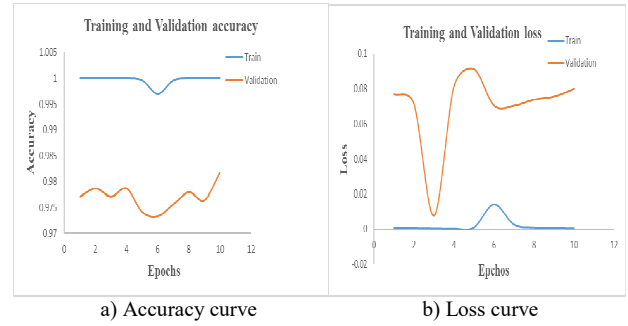


Fig. 13: Bi-LSTM Training and Validation curve

E. Model Evaluation

Performance evaluation is done using the metrics such as an accuracy, precision, recall and F-1 score [10]. Description of these parameters are concisely explained as below:

i. Accuracy: Accuracy is the ratio of the number of accurate predictions to the total number of input predictions.

$$Accuracy = \frac{tr.p + tr.n}{tr.p + tr.n + fa.n + fa.p} \quad (1)$$

ii. Precision: The precision is calculated as a ratio of true positives and false positives to all positive forecasts, which includes genuine positive predictions.

$$Precision = \frac{tr.p}{tr.p + fa.p} \quad (2)$$

iii. Recall: Out of all actual positive predictions, recall calculates the proportion of positive predictions that are correctly categorized.

$$Recall = \frac{tr.p}{tr.p + fa.n} \quad (3)$$

iv. F1-Score: It is used as the Harmonic average of precision and recall.

$$F1score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

where fa.n stands for false negative, fa.p for false positive, tr.p for true positive, and tr.n for true negative. Accuracy metric used here is to find how correctly the models are used to categorize whether the news is fake or true. If the accuracy is higher, then the model is expected to classify the true and fake news accurately. The accuracy of the model LSTM is 97.57% and that of Bi-directional LSTM is 98.15% as shown in Fig.14.

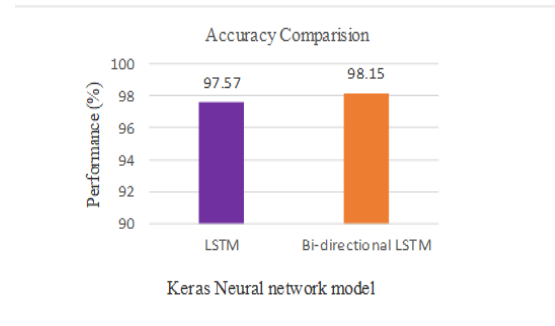


Fig. 14: Accuracy of two Keras models.

F. Experiment results of political based dataset

When the LSTM model is applied to the political news dataset, the Precision, Recall, and F1-score are 99%,

97%, and 98% for the fake label and 96%, 98%, and 97% for the real label, respectively. For the bi-directional LSTM model, the results are as follows: precision, recall, and F-1 score are 99%, 97%, and 98% for fake labels and 97%,99%,98% for real labels, respectively as depicted in the TABLE II

TABLE II CLASSIFICATION REPORT OF POLITICAL NEWS DATASET.

Classifier	Label	Precision	Recall	F1-Score
LSTM	Fake	0.99	0.97	0.98
	Real	0.96	0.98	0.97
Bi-directional LSTM	Fake	0.99	0.97	0.98
	Real	0.97	0.99	0.98

The experimental study discloses that the proposed method offers a better accuracy of about 98.15% on Bi-LSTM and 97.57% on LSTM when compared with Kera's neural network model trained with N-gram or sequence vectors on GitHub repository which is 90.3% of accuracy [2]. In another work [14], Bidirectional LSTM model is used to detect whether the news is fake or real on websites and newspaper data achieving overall accuracy of 84%, and proposed method achieved a better accuracy of 98.15% on similar model.

V. CONCLUSION AND FUTURE SCOPE

The purpose of this work is accurate classification of political news articles as Fake or True using deep learning models. The link between the headline and body of a news article is depicted in the suggested model using one-hot encoding. The possibility that the headline's terms will also appear in the article's body language weighs the veracity of the story. When pre-processing data, stop-words, Lemmatization, and tokenization are used to pre-process the input data. At last, the extracted features are fed for training into the Kera's neural network models such as LSTM, Bidirectional LSTM with an ADAM optimizer. The analysis reveals that LSTM obtains an accuracy of 97.57% and that on Bi-directional LSTM is 98.15%. The results show Bi-directional LSTM model is more effective than the LSTM model. In future, research work can be applied to the society in detecting the propagation of fake news related to text data along with semantic metadata on the images using other DL approaches like CNN, RNN etc. and to other datasets like COVID-19 tweets, entertainment, education related posts.

REFERENCES

- [1]. S. H. Kong, L. M. Tan, K. H. Gan and N. H. Samsudin, "Fake News Detection using Deep Learning," *2020 IEEE 10th Symposium on Computer Applications & Industrial Electronics (ISCAIE)*, 2020, pp. 102-107.
- [2]. C. K. Hiramath and G. C. Deshpande, "Fake News Detection Using Deep Learning Techniques," *2019 1st International Conference on Advances in Information Technology (ICAIT)*, 2019, pp. 411-415.
- [3]. H. Saleh, A. Alharbi and S. H. Alsamhi, "OPCNN-FAKE: Optimized Convolutional Neural Network for Fake News Detection," in *IEEE Access*, vol. 9, pp. 129471-129489, 2021.
- [4]. Kai Shu, Amy Sliva, Suhan Wang, Jiliang Tang, and Huan Liu, "Fake News Detection on Social Media: A Data Mining Perspective", *ACM SIGKDD Explorations Newsletter Volume 19 Issue 1*, June 2017 pp 22–36.
- [5]. Dheeraj Kumar Dixit, Amit Bhagat, Dharmendra Dangi, "Automating fake news detection using PPCA and levy flight-based LSTM", *Springer Nature in APPLICATION OF SOFT COMPUTING*, pp: 12545–12557, 2022.
- [6]. I. Kadek Sastrawan, I.P.A. Bayupati, Dewa Made Sri Arsa, "Detection of fake news using deep learning CNN-RNN based methods", *Elsevier in The Korean Institute of Communications and Information Sciences*, pp: 396-408, 2021.
- [7]. M. Umer, Z. Imtiaz, S. Ullah, A. Mehmood, G. S. Choi and B. -W. On, "Fake News Stance Detection Using Deep Learning Architecture (CNN-LSTM)," in *IEEE Access*, vol. 8, pp. 156695-156706, 2020.
- [8]. V. -I. Ilie, C. -O. Truică, E. -S. Apostol and A. Paschke, "Context-Aware Misinformation Detection: A Benchmark of Deep Learning Architectures Using Word Embeddings," in *IEEE Access*, vol. 9, pp. 162122-162146, 2021.
- [9]. A. Kumar, J. P. Singh, and A. K. Singh, "Covid-19 fake news detection using ensemble-based deep learning model," *IT Professional*, vol. 24, no. 2, pp. 32–37, 2022
- [10]. M. F. Mridha, A. J. Keya, M. A. Hamid, M. M. Monowar and M. S. Rahman, "A Comprehensive Review on Fake News Detection with Deep Learning," in *IEEE Access*, vol. 9, pp. 156151-156170, 2021.
- [11]. Jamal Abdul Nasir, Osama Subhani Khan, Iraklis Varlamis,, "Fake news detection: A hybrid CNN-RNN based deep learning approach", *International Journal of Information Management Data Insights*, Vol. 1, Issue 1, 2021, ISSN 2667-0968.
- [12]. Kaliyar, R.K., Goswami, A. & Narang, P, "EchoFakeD: improving fake news detection in social media with an efficient deep neural network", *ACM in Neural Computing and Applications*, vol. 33, Issue 14, pp:8597–8613, 2021.
- [13]. G. Sansonetti, F. Gaspiretti, G. D'aniello and A. Micarelli, "Unreliable Users Detection in Social Media: Deep Learning Techniques for Automatic Detection," in *IEEE Access*, vol. 8, pp. 213154-213167, 2020.
- [14]. T. Islam, M. A. Hosen, A. Mony, M. T. Hasan, I. Jahan and A. Kundu, "A Proposed Bi-LSTM Method to Fake News Detection," *2022 International Conference for Advancement in Technology (ICONAT)*, Goa, India, 2022, pp. 1-5.
- [15]. R. Katarya, D. Dahiya, S. Checker et al., "Fake news detection system using featured-based optimized msvm classification," *IEEE Access*, vol. 10, pp. 113 184–113 199, 2022.
- [16]. Z. Zhang, "Improved Adam Optimizer for Deep Neural Networks," *2018 IEEE/ACM 26th International Symposium on Quality of Service (IWQoS)*, IEEE, 2018, pp. 1-2.